

Technical Design Document: ZachariahKersey_ProgrammingExercise_CSV

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Program Description: Gathers the records of all students and saves to a CSV file, then display the values in table

Function: gatherInputs

Gather the total number of students, their names and their exam grades

Parameters:

None

Variables:

students (list): A list of students records

numberOfStudents (int): Total number of students to keep records for

firstName (str): First name of the student

lastName (str): Last name of the student

gradeOne(int): First exam grade

gradeTwo(int): Second exam grade

gradeThree(int): Third exam grade

Logic:

1. Gather the number of students to keep records for, handling input errors
2. Loop through adding the names and exam grades for the number of students to the student records, handling input errors
3. Run the writeToCSV function, passing the list of student records

Returns:

None

Function: readGrades

Open the grades from a CSV file and display them in a table format

Parameters:

None

Variables:

reader (list): A list of header row

Logic:

1. Open the grades.csv file
2. Print the header of the CSV file
3. Print all the rows of values from the CSV file

Returns:

None

Function: writeToCSV

Gather the total number of students, their names and their exam grades

Parameters:

students (list): A list of students records

Variables:

HEADER (list): A list of header row

Logic:

1. Open the grades.csv file
2. Write header row and students records to the open grades.csv file

Returns:

None

#Link to your repository: <https://github.com/Z-Money?tab=repositories>

```
"C:\Users\zrun2\OneDrive\Education\2025-2026 12TH Grade\4th COP 2373 - Melichar\.venv\Scripts\python.exe" "C:\Users\zrun2\OneDrive\Education\2025-2026 12TH Grade\4th COP 2373 - Melichar\grades.py"
How many students would you like to enter grades for: 3
Enter student 1's first name: Zachariah
Enter student 1's last name: Kersey
Enter student 1's first exam grade: 98
Enter student 1's second exam grade: 85
Enter student 1's third exam grade: 93
Enter student 2's first name: Joseph
Enter student 2's last name: Barr
Enter student 2's first exam grade: 100
Enter student 2's second exam grade: 84
Enter student 2's third exam grade: 94
Enter student 3's first name: Alexa
Enter student 3's last name: Jaime
Enter student 3's first exam grade: 99
Enter student 3's second exam grade: 78
Enter student 3's third exam grade: 92

Data successfully written to grades.csv.

First Name    Last Name    Exam 1    Exam 2    Exam 3
Zachariah     Kersey       98        85        93
Joseph        Barr         100       84        94
Alexa         Jaime        99        78        92

Process finished with exit code 0
```

```
"C:\Users\zrun2\OneDrive\Education\2025-2026 12TH Grade\4th COP 2373 - Melichar\.venv\Scripts\python.exe" "C:\Users\zrun2\OneDrive\Education\2025-2026 12TH Grade\4th COP 2373 - Melichar\grades.py"
How many students would you like to enter grades for: John
Please enter an integer value for the number of students.
How many students would you like to enter grades for: 2
Enter student 1's first name: John
Enter student 1's last name: Franklin
Enter student 1's first exam grade: Jr
Enter student 1's second exam grade: 22
Enter student 1's third exam grade: 33
Please enter an integer value for their exam grades.
Enter student 1's first exam grade: |
```